

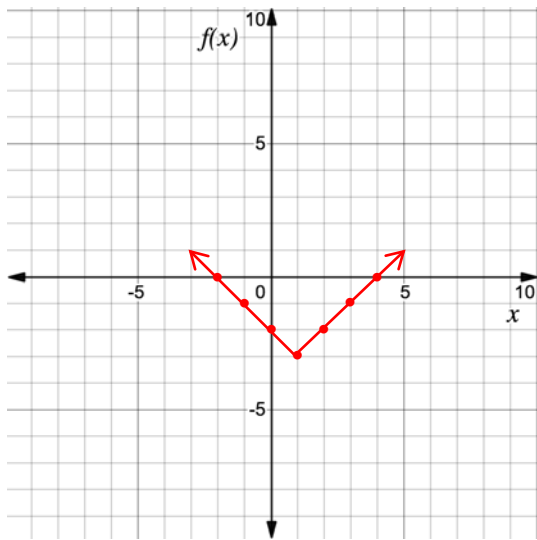
Algebra 1
Unit 10
Lesson 1 Practice

Name _____
Date _____
Period _____

1. Complete the table of values for $f(x) = |x - 1| - 3$.

x	-2	-1	0	1	2	3	4
$f(x)$	0	-1	-2	-3	-2	-1	0

2. Use the table of values to graph $f(x) = |x - 1| - 3$ on the coordinate grid.



3. What is the vertex of $f(x) = |x - 1| - 3$?
 $(1, -3)$

4. What is the equation for the axis of symmetry of $f(x) = |x - 1| - 3$?
 $x = 1$

5. What are the x and y -intercept(s) of $f(x) = |x - 1| - 3$?
 x -int: $(-2, 0)(4, 0)$
 y -int: $(0, -2)$

6. What are the domain and range of $f(x) = |x - 1| - 3$?
Domain: All real numbers
Range: $y \geq -3$

7. Identify the intervals on which the function $f(x) = |x - 1| - 3$ is increasing and/or decreasing.

Increasing: $1 < x < \infty$

Decreasing: $-\infty < x < 1$

8. Identify the intervals on which the function $f(x) = |x - 1| - 3$ is positive and negative.

Positive: $x < -2$ or $x > 4$

Negative: $-2 < x < 4$

For questions 9-10, consider the absolute value function $f(x) = |x + 1| - 4$.

9. Complete the table of values for $f(x) = |x + 1| - 4$.

x	-4	-3	-2	-1	0	1	2
$f(x)$	-1	-2	-3	-4	-3	-2	-1

10. Use the table of values to graph $f(x) = |x + 1| - 4$ on the coordinate grid.

